Problem Statement:

Predict the profit based on the investment in each department

Using Decision Tree Algorithm

SL No	Criterion	Splitter	Max Features	R^2 Value
1	squared_error	best	None	0.89868884
2	friedman_mse	best	None	0.92160242
3	absolute_error	best	None	0.92848658
4	poisson	best	None	0.92940904
5	squared_error	random	None	0.8558754
6	friedman_mse	random	None	0.93687294
7	absolute_error	random	None	0.76675981
8	poisson	random	None	0.86335449
9	squared_error	best	100	0.88519854
10	friedman_mse	best	100	0.92944234
11	absolute_error	best	100	0.93121715
12	poisson	best	100	0.93426023
13	squared_error	best	0.9	0.7699807
14	friedman_mse	best	0.9	0.93037168
15	absolute_error	best	0.9	0.80008462
16	poisson	best	0.9	0.83269058
17	squared_error	best	sqrt	0.79617358
18	friedman_mse	best	sqrt	0.78361735
19	absolute_error	best	sqrt	0.48433558
20	poisson	best	sqrt	0.65409334
21	squared_error	best	log2	0.62763333
22	friedman_mse	best	log2	0.15587832
23	absolute_error	best	log2	0.41765639
24	poisson	best	log2	0.44262552
25	squared_error	random	100	0.86198013
26	friedman_mse	random	100	0.79674022
27	absolute_error	random	100	0.93053113
28	poisson	random	100	0.73221932
29	squared_error	random	0.9	0.4450546
30	friedman_mse	random	0.9	0.8408396
31	absolute_error	random	0.9	0.96634141
32	poisson	random	0.9	0.62360167
33	squared_error	random	sqrt	0.09375633
34	friedman_mse	random	sqrt	0.45115096
35	absolute_error	random	sqrt	0.09939228

36	poisson	random	sqrt	0.20964794
37	squared_error	random	log2	0.81367716
38	friedman_mse	random	log2	-0.1175284
39	absolute_error	random	log2	0.77966747
40	poisson	random	log2	0.92530792